

FNB Sales Case Study Analysis

Comprehensive Project Description

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Executive Summary

The FNB Sales Case Study Analysis is an end-to-end retail sales analytics project completed to demonstrate practical data analytics skills across the complete analytics lifecycle. The project involved cleaning, transforming, analysing, and visualising historical retail sales data to generate meaningful business insights and decision-support information.

Using a dataset containing 1,057 daily sales records spanning 30 December 2013 to 16 November 2016, the analysis focused on measuring sales performance, profitability, pricing behaviour, promotional effectiveness, and customer purchasing patterns.

The project combined multiple industry-standard analytics tools including Databricks, SQL, Power BI, Microsoft Excel, Draw.io, Looker Studio, and Lovable to simulate a real-world business intelligence solution.

Project Objectives

Analyse historical retail sales performance; calculate key financial and operational metrics; measure profitability; identify promotional periods; calculate price elasticity of demand; discover seasonal and weekly sales trends; build interactive dashboards; present findings through business intelligence reports.

Dataset: 1,057 records | Date, Sales, Cost of Sales, Quantity Sold | 30 Dec 2013–16 Nov 2016.

Business questions covered revenue, seasonality, profitability, pricing, promotions and elasticity.

Key metrics: Average Unit Price R37.25; Gross Profit -3.81%; Gross Profit Per Unit -0.88%; Elasticity Dec 2014 -1.85; Jul 2015 -2.10; Jan 2016 -1.45; Average Elasticity -1.80.

Key findings: Revenue R186.91M; Units Sold 5.28M; Best Day Saturday; Best Month December; Best Quarter Q4; Product classified as Price Elastic.

Technologies Used

Databricks, SQL, Power BI, Microsoft Excel, Draw.io, Looker Studio and Lovable were used for data processing, SQL analysis, dashboard development, project planning, reporting and presentation.

Repository Structure

README, data, sql, databricks, powerbi, excel, looker_studio, drawio, lovable, documentation and images folders.

Skills Demonstrated

Business Intelligence, SQL, Spark SQL, Databricks, Power BI, Excel, Dashboard Design, KPI Development, Data Visualisation, Statistical Analysis, Price Elasticity Analysis, Financial Metrics and Business Reporting.

Project Outcome

The project demonstrates an end-to-end analytics workflow from raw data to executive dashboards and business recommendations, showcasing practical analytical and business intelligence skills.